CAMPUS TRANSPORT ALLOCATION

1. Algorithm

1. Start the program.

2. Display the program title.

3. Read the number of routes.

4. Repeat the following steps for each route:

a. Read the route name, number of passengers, and bus capacity.

b. If capacity is less than or equal to zero, or passengers are negative, display an invalid-input message and continue to the next route.

c. Calculate buses required using: (passengers + capacity - 1) // capacity.

d. Calculate passengers served using: min(passengers, buses × capacity).

e. Calculate unserved passengers = passengers – passengers served.

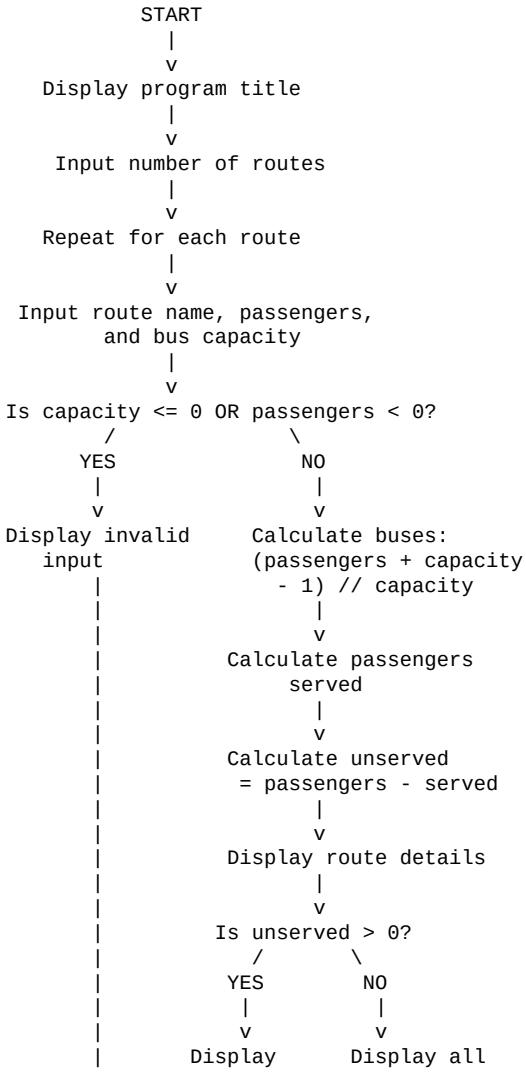
f. Display route details, buses required, passengers served, and unserved passengers.

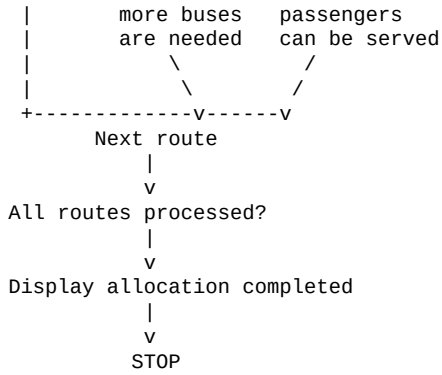
g. If unserved passengers are greater than zero, display an alert that more buses are needed; otherwise, display that all passengers can be served.

5. After processing all routes, display “Transport allocation completed!”.

6. Stop the program.

2. Flowchart




more buses
are needed
passengers
can be served
Next route
All routes processed?
Display allocation completed
STOP

3. Pseudocode

```
BEGIN

    PRINT "CAMPUS TRANSPORT ALLOCATION"

    INPUT routes

    FOR i = 1 TO routes DO

        PRINT "Route", i

        INPUT name
        INPUT passengers
        INPUT capacity

        IF capacity <= 0 OR passengers < 0 THEN
            PRINT "Invalid input! Enter valid values."
            CONTINUE
        END IF

        buses = (passengers + capacity - 1) // capacity

        served = MIN(passengers, buses * capacity)

        unserved = passengers - served

        PRINT route name
        PRINT total passengers
        PRINT bus capacity
        PRINT buses required
        PRINT passengers served
        PRINT unserved passengers

        IF unserved > 0 THEN
            PRINT "Alert: More buses are needed!"
        ELSE
            PRINT "All passengers can be served."
        END IF

    END FOR

    PRINT "Transport allocation completed!"

END
```

4. Sample Output

```
CAMPUS TRANSPORT ALLOCATION
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Enter number of routes: 1

Route 1
Enter route name: Main Gate
Enter number of passengers: 85
Enter bus capacity: 40

--- Route Details ---
Route: Main Gate
Total passengers: 85
Bus capacity: 40
Buses required: 3
Passengers served: 85
Unserved passengers: 0
All passengers can be served.

Transport allocation completed!
```

Note

The sample follows the uploaded Python program. Since the program calculates enough buses to cover the passenger count, the unserved-passenger value is zero for valid non-negative passenger counts and positive bus capacities.